Strategic HDB Flat Selection: Lease, Location & Affordability Insights

Business analysis, suited to every client's needs.
Prepared by – **Ng Yu Hang**

Business Objective

To help new and prospective HDB resale flat buyers in Singapore make informed purchasing decisions by analyzing and visualizing key factors such as:

- **Affordability** (price per sqm, overall flat pricing)
- **Remaining lease years** (lease risk and long-term value)
- **MRT accessibility** (location convenience and connectivity)
- **Flat type and size** (suitability for different buyer profiles)
- This objective supports a **data-informed advisory framework** for different buyer needs (e.g., families, young buyers, investors).

Prepared by: **Ng Yu Hang – Property Agent Specialist**

Data set dictionary – used columns only

Given dataset:

Column Name	Description
month	Month and year of the resale transaction
town	HDB town where the flat is located
flat_type	Type of flat (e.g., 3 ROOM, 4 ROOM)
storey_range	General floor level range of the flat
floor_area_sqm	Size of the flat in square meters
remaining_lease	Estimated number of years left on the lease
resale_price	Total resale price paid for the flat

Supplementary dataset:

Column Name	Description
building	Building name of the flat
postal	Postal code of the building
floor_area_sqft	Floor area of the flat in square feet
price_per_sqft	Resale price divided by floor area in sqft
distance_to_mrt	Distance from flat to nearest MRT station (in meters)
distance_to_school	Distance from flat to nearest primary school (in meters)
latitude	Latitude coordinate of the flat
longitude	Longitude coordinate of the flat
planning_area	URA planning area the flat belongs to
region	URA-defined region (e.g., Central, West)
nearest_mrt_line_color	Color of the MRT line closest to the flat

Data Wrangling process

- ✓ **Cleaned and standardized flat_type (e.g., converted “3 RM” to “3 ROOM”, removed inconsistent casing and spacing)**
- ✓ **Standardized street_name using uppercase conversion, whitespace removal, and abbreviation expansion (e.g., “ST.” → “STREET”) – Consistent naming throughout the column.**
- ✓ **Processed storey_range to extract the lower bound floor number for analysis (e.g., "04 TO 06" → 4)**
- ✓ **Converted resale_price and floor_area_sqm to numeric, stripping invalid characters and coercing errors**
- ✓ **Parsed remaining_lease from messy text into clean numeric years (e.g., “91 years 6 months” → 91.5)**
- ✓ **Detected and corrected a known resale price outlier (~$7 million) to prevent skewed analysis (changed to 700k based on resale prices of the same building)**
- ✓ **Identified unrealistic floor area values and flagged them as potential outliers using plotly.**
- ✓ **Ensured uniform formatting for town names (e.g., “SENG-KANG” → “SENGKANG”) and merged aliases**
- ✓ **Dropped irrelevant or corrupted records and ensured all key fields are complete and usable.**

- 303 Duplicate Rows are removed

Original Dataset rows: 202,764 Rows
Cleaned dataset rows: 202,461 Rows

- New dataset was already cleaned, merged on Town by Relationship

Ensures no duplicated rows when there are multiple entries of same data (e.g. multiple nearby MRTs, etc)

Chart 1 – MRT Accessibility across price-tiered towns

Chart 1 – MRT-Accessible Flats by Price Tier

Objective: Help home buyers identify towns offering affordable resale flats with strong MRT accessibility.

What This Shows:

•Towns are colored by average price per sqm (Green = more affordable, Red = more expensive).

•Flat icons are **sized** by MRT accessibility score and **colored** by nearest MRT line.

Key Insights:

•Affordable and MRT-accessible flats are concentrated in certain regions (e.g., North and West).
•Buyers can visually compare which towns balance both low cost and high accessibility.
•Useful for filtering resale options based on transport convenience and budget suitability.

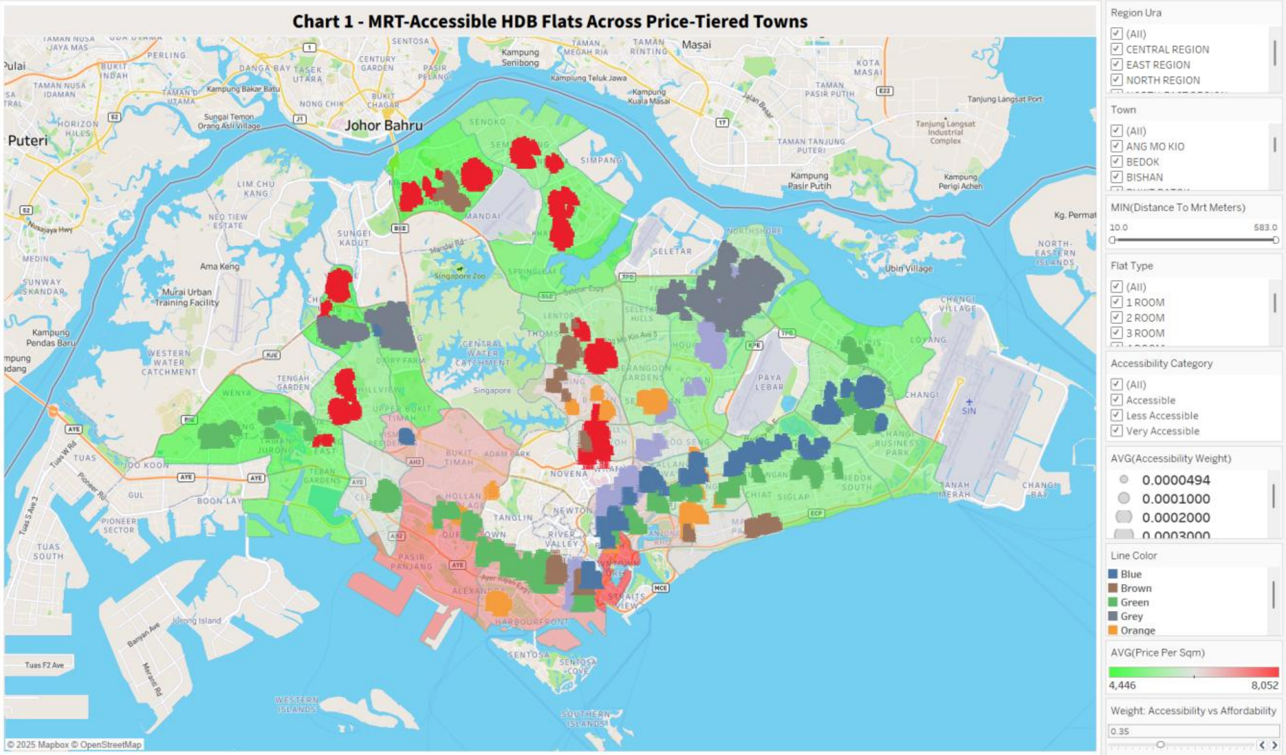

Chart 2 – Quadrant Analysis: Floor Area vs School Proximity

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Objective: Identify flats that are both spacious and close to primary schools—ideal for family-focused buyers.

What This Shows:

•Quadrants classify flats based on **size (sqm)** and **distance to nearest school (m)**.

•Top-left (blue): *Ideal* — large flats within walkable distance to schools.

•Bottom-right (red): small flats far from schools — *least desirable* for families.

Key Insights:

•Ideal flats (spacious & walkable) are scarce and clustered.

•Filters allow users to adjust definitions of "spacious" and "near" based on their needs.

•Helps prioritize family-friendly resale options in limited locations.

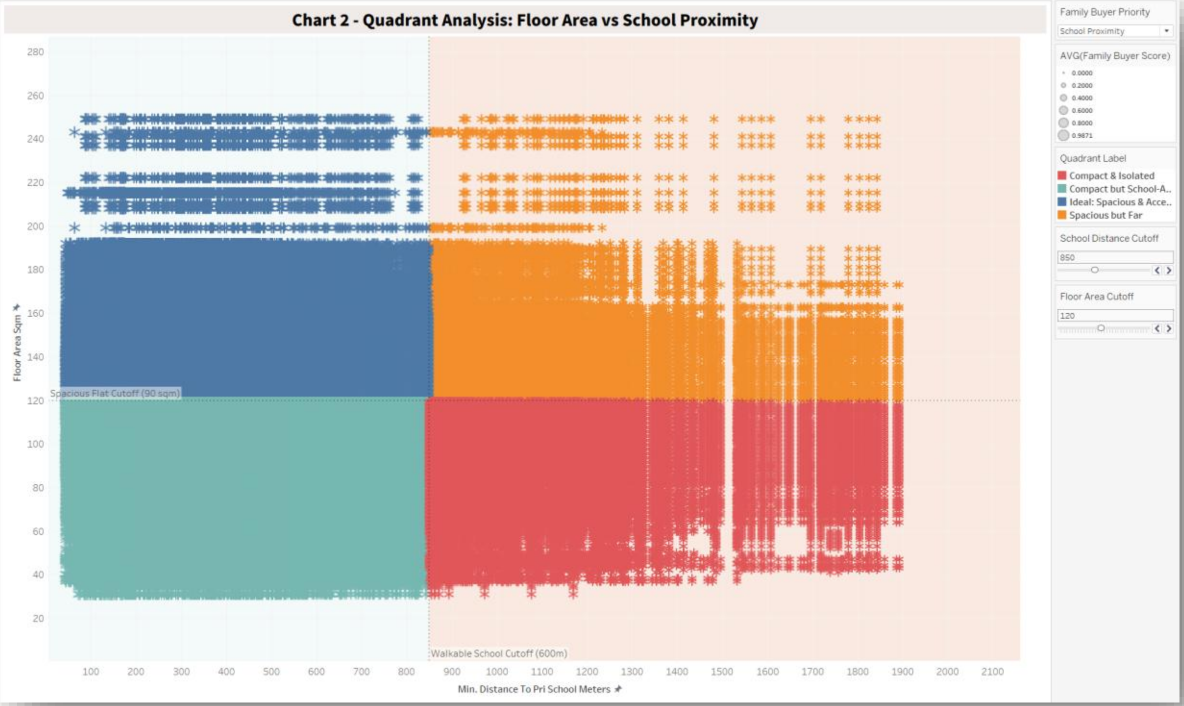


Chart 3 – Yearly Resale Affordability Classification (Threshold: $500K)
Chart 4 – Buyer Suitability Score: <Buyer Focus>

Chart 3 – Yearly Resale Affordability Classification (Threshold: $500K)

Objective: Track flat types that remain affordable across years for budget-conscious buyers.

What This Shows:
- ✓ = Majority of transactions under $500K for that year and flat type
- × = Mostly unaffordable
- Colors reflect affordability consistency:
  - Green = Always affordable
  - Yellow = Sometimes affordable
  - Red = Consistently unaffordable

Key Insights:
- 1–3 Room flats are reliably affordable across all years.
- 4 Room flats became unaffordable after 2020.
- Larger flat types (5 Room, Executive, Multi-Gen) exceed $500K every year.
- Budget buyers should focus on smaller flat types to stay within limit.

Chart 4 – Buyer Suitability Score (Dynamic by Buyer Focus)

Objective: Rank towns based on suitability for different buyer types (Family, Budget, Investor, Balanced).

What This Shows:
- A weighted score is calculated per town, based on the selected **Buyer Focus**.
- Users can switch between profiles (e.g., Family-Oriented, Budget-Conscious) to reveal different rankings.
- Color gradient (Red → Green) highlights suitability strength relative to others.

Key Insights:
- Town rankings shift depending on buyer needs—e.g., Pasir Ris ranks high for families, but may differ for investors.
- Enables personalized decision-making by aligning town selection with buyer goals and constraints.

Town	
CENTRAL AREA	1,257.8
QUEENSTOWN	1,272.5
BUKIT MERAH	1,342.2
BUKIT TIMAH	1,665.8
KALLANG/WHAMPOA	1,332.2
BISHAN	1,650.1
MARINE PARADE	1,322.3
CLEMENTI	1,330.8
TOA PAYOH	1,327.6
GEYLANG	1,284.7
PUNGGOL	1,478.5
SERANGOON	1,555.8
ANG MO KIO	1,308.2
TAMPINES	1,598.9
SENGKANG	1,531.7
BEDOK	1,380.8
BUKIT BATOK	1,451.4
HOUGANG	1,537.9
SEMBAWANG	1,527.3
BUKIT PANJANG	1,572.2
JURONG EAST	1,494.7
YISHUN	1,428.8
PASIR RIS	1,837.6
JURONG WEST	1,561.1
CHOA CHU KANG	1,662.5
WOODLANDS	1,624.4

Chart 3 – Yearly Resale Affordability Classification (Threshold: $500K)
Chart 4 – Buyer Suitability Score: <Buyer Focus>

Chart 5 – Which Flat Types Pose Lease Risks? (CPF Cutoff Highlighted)

Objective: Evaluate which flat types are at risk of falling below the CPF eligibility lease threshold (60 years remaining).

What This Shows:

•Boxplots display lease distribution for each flat type.
•The dashed line marks the CPF financing cutoff at 60 years.
•Flats below this line may face loan and grant restrictions.

Key Insights:

•**1 ROOM and Multi-Generation** flats have lower median leases, often below CPF eligibility.
•**2–4 ROOM flats** have wide lease variance; some may still pose CPF risks.
•**5 ROOM and Executive flats** are generally safer but should still be checked individually.
•Useful for avoiding flats that limit financing or resale appeal due to lease decay.

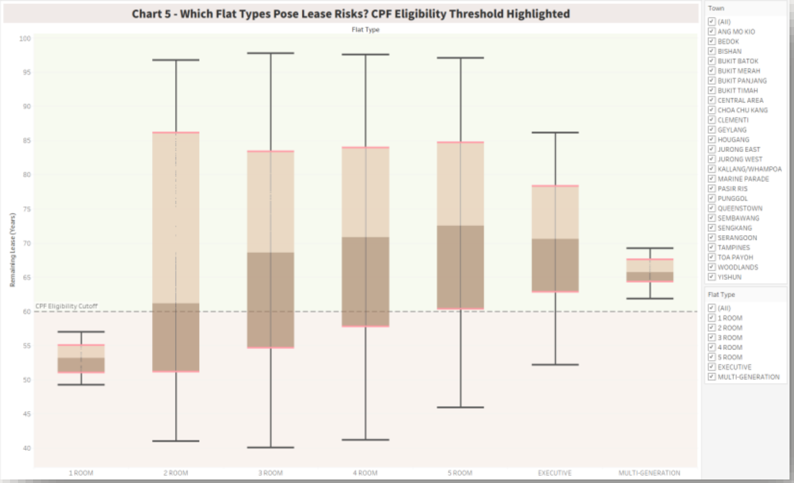


Chart 6 – Quadrant Analysis: Lease Duration vs Price Efficiency

Objective: Help buyers identify flats that offer long leases at affordable prices while avoiding lease-risk or overpricing.

What This Shows:

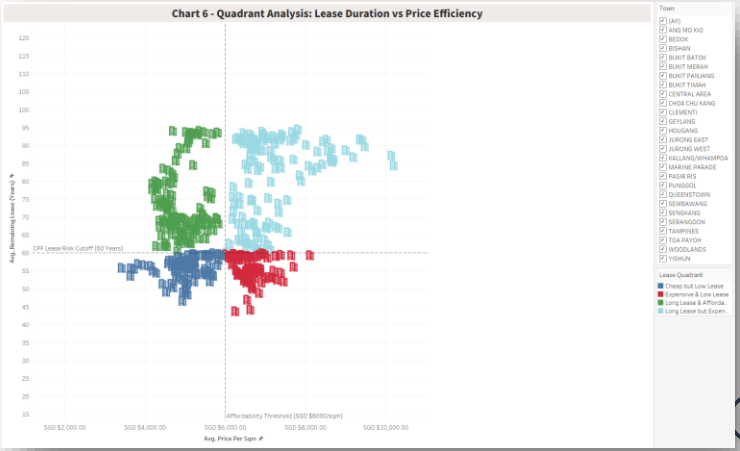
•X-axis: Price per sqm (affordability threshold at $6,000)
•Y-axis: Avg. remaining lease (CPF risk at 60 years)

Quadrants:

- **Green:** Long lease & affordable -> best value
- **Light Blue:** Long lease but expensive
- **Dark Blue:** Cheap but lease risky
- **Red:** Expensive and lease risky -> Avoid

Key Insights:

•Optimal flats are clustered in the **green quadrant**, offering strong value and financing eligibility.
•Flats in the **red quadrant** are both costly and risk CPF financing limitations.
•A quick visual tool to balance budget and lease security in flat selection.



Dashboard 1:

Strategic Flat Selection: MRT, Schools, Space and Resale Affordability

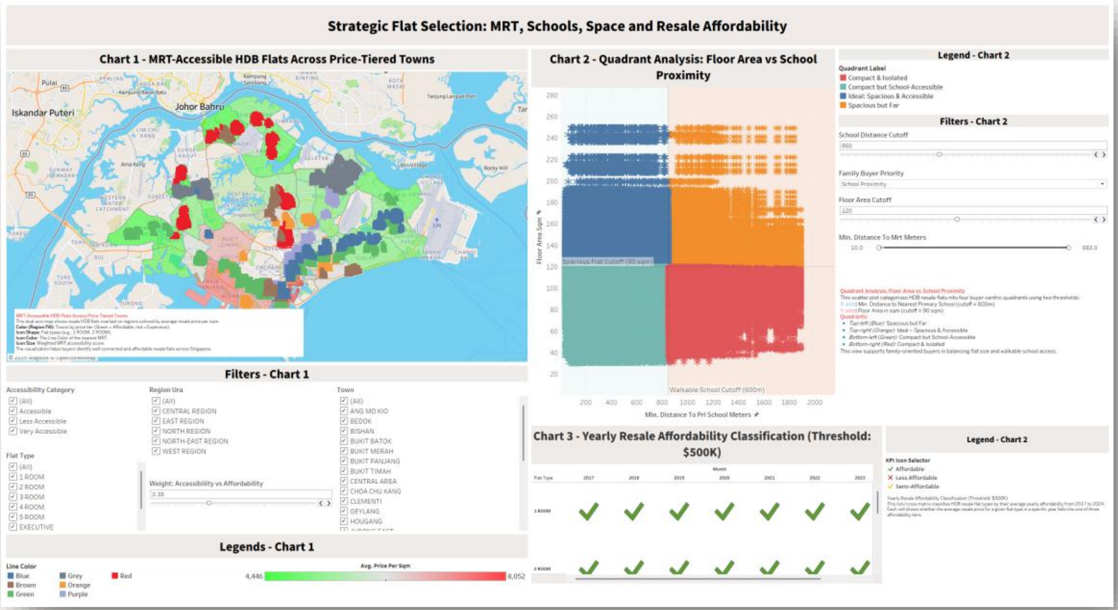
Objective: ***Guide new buyers to affordable, accessible resale flats with good space and proximity to schools.***

Chart 1: MRT-accessible flats mapped by affordability tiers
→ Green towns offer cheapest price per sqm with MRT access.
→ Icon size reflects MRT accessibility; icon color reflects MRT line.

Chart 2: Quadrant analysis of flat size vs school proximity
→ Large flats near schools (top-left quadrant) are rare and valuable.
→ Filter by family priorities to identify ideal segments.

Chart 3: Yearly trend of affordable flats (below $500K)
→ Consistent availability across years for multiple towns.

Key Insight: Buyers can prioritize town, space, MRT access, and school proximity using interactive filters to find ideal flats under budget.



Dashboard 2:

Strategic Flat Selection by Buyer Profile: Suitability, Lease Risks, and Value Zones

Objective: ***Help buyers align flat choices with their priorities - family suitability, CPF lease eligibility, and price efficiency.***

Chart 4 – Buyer Suitability Score

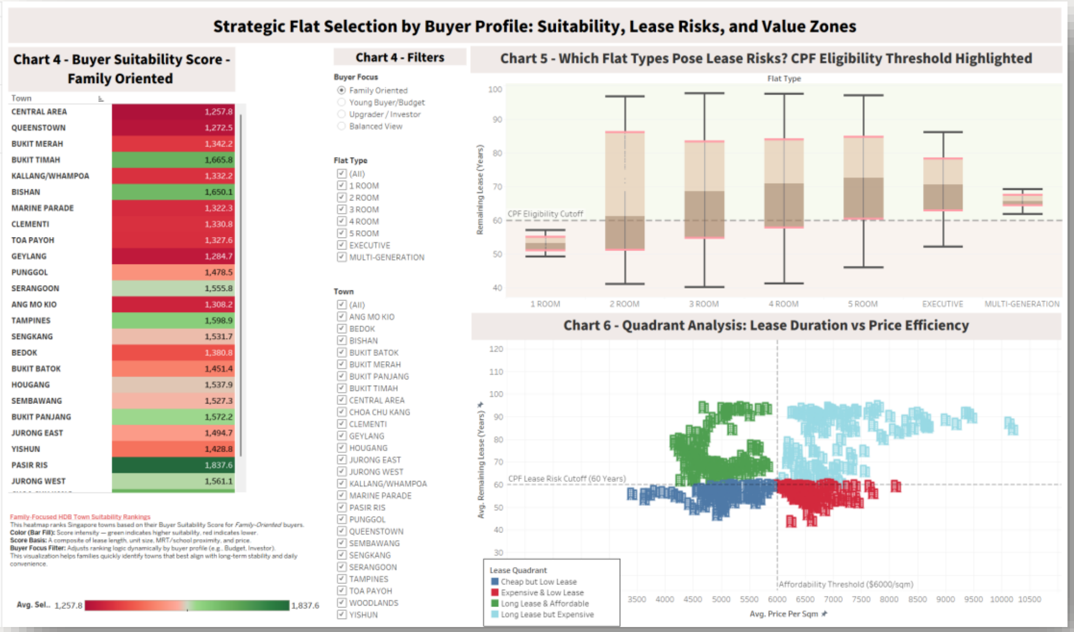
→ Ranks towns by family-oriented priorities (lease, space, MRT, schools).
→ Green towns offer highest suitability.
→ Buyer Focus filter dynamically adjusts scores by buyer type.

Chart 5 – Lease Risk by Flat Type

→ Boxplots show lease ranges per flat type.
→ CPF cutoff at 60 years highlighted.
→ 1 ROOM and Multi-Gen flats most at risk; 3–5 ROOM flats generally safer.

Chart 6 – Lease Duration vs Price Efficiency

→ Quadrants compare price per sqm against average lease.
→ Green = Best value (long lease, affordable).
→ Red = Risky (expensive, short lease).



Key Insight:

This dashboard helps buyers avoid lease-risk flats, compare town suitability, and balance price vs lease value - all filtered by buyer goals.

Suggestion:

Pasir Ris stands out as a strong overall choice for most buyers.
It offers a long average lease, strong MRT accessibility, and ranks among the top for family suitability. Despite being located further from the city core, it provides high value for budget-conscious and family-oriented buyers looking for affordable, spacious flats near schools.

< 1km

Away from the nearest MRT (Pasir Ris) for all flats

< 900m

Away from the neaarest Primary School for all flats

> 60 yrs lease

For 99% of all flat types in Pasir Ris

Reference link (new dataset):
https://www.kaggle.com/datasets/lzytim/hdb-resale-prices